

Observable Bounds on Ontological Dimension: A Constructive Consequence of Projection Fiber Theory

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Abstract

We derive a computable lower bound on the dimension of ontological state spaces from observable independence results. Given a non-injective projection $\Pi : H_{\text{ont}} \rightarrow H_{\text{obs}}$ and n predicates that are independently non-invariant on the fibers of Π , the fiber cardinality satisfies $|\text{Fib}(o)| \geq 2^n$. Each independent predicate certifies a binary split that the projection cannot collapse. The bound is observable: independence is detected from H_{obs} alone using the invariance criterion of Paper I, requiring only the projection map and an SMT solver. We verify this constructively in Kleis: two independent predicates over the Cantor shadow theory (the Continuum Hypothesis and the Generalized Continuum Hypothesis at $\aleph_1$) force at least four ontologically distinct models in the same projection fiber, all confirmed by Z3. A second, independent bound arises from fiber dynamics: the orbit length of a fiber-preserving evolution operator certifies fiber cardinality from below. The combined bound takes the maximum. This result inverts the epistemic direction of Projected Ontology Theory: Paper I established what cannot be known from the projection side (the specific action, the thermodynamic arrow, the flow character); this paper establishes what must exist — a minimum complexity of the hidden layer, computable from observational data. The projection determines not the content of H_{ont} , but a lower bound on its complexity. All 33 verification results for the extended `pot_bridge.kleis` are machine-checked and reproducible.

Keywords: fiber dimension, ontological degrees of freedom, projection fibers, independence counting, lower bounds, formal verification, Continuum Hypothesis, Generalized Continuum Hypothesis

1 Introduction

In a companion paper [1], we introduced a geometric framework for detecting logical independence: a statement S is independent of a projected theory if and only if S is non-invariant on the fibers of the projection map Π . We demonstrated this computationally by detecting the independence of the Continuum Hypothesis from Cantor’s cardinal arithmetic using Z3, and extended the framework to show that the fibers carry intrinsic structure — geometry, dynamics, and admissible selection principles — while proving that the specific variational principle governing the ontological layer is epistemically inaccessible from the projection side.

That paper was primarily negative in character: it characterized what cannot be known from H_{obs} . The epistemic boundary, the flow underdetermination, the arrow underdetermination — all establish limits on observational knowledge.

This paper asks the complementary question: given observations alone, what can we certify about the ontological layer? Specifically, can we set a minimum on the number of ontological degrees of freedom?

We answer affirmatively. The key insight is that each logically independent predicate detected from the observation side certifies a binary split in the fiber. If n predicates are pairwise independently non-invariant on the fibers of Π , then the fiber must contain at least 2^n distinct points to realize all truth-value combinations. This is the Fiber Dimension Theorem:

$$|\text{Fib}(o)| \geq 2^n$$

where n is the number of detected independent predicates. The bound is computable because independence detection uses the invariance criterion of Paper I: run the SMT solver on $P(m_1) \wedge \neg P(m_2)$ with $\Pi(m_1) = \Pi(m_2)$. Each satisfiable result is a structural certificate that doubles the lower bound.

A second, independent source of lower bounds comes from fiber dynamics: if the fiber-preserving evolution operator `fiber_evolve` produces k consecutive distinct states, then $|\text{Fib}(o)| \geq k$. The combined bound is the maximum of both estimates.

For the Cantor instantiation: the Continuum Hypothesis (CH) gives one independent predicate ($n \geq 1$, so $|\text{Fib}| \geq 2$). The Generalized Continuum Hypothesis at $\aleph_1$ (GCH) provides a second, orthogonal predicate ($n \geq 2$, so $|\text{Fib}| \geq 4$). The fiber dynamics orbit gives $k \geq 2$. Combined: $|\text{Fib}| \geq \max(4, 2) = 4$, verified by Z3 in under 8 seconds.

The philosophical significance is precise: the projection cannot tell you WHAT is in the fiber, but it CAN tell you HOW MUCH must be there. Every independence result detected from H_{obs} is a structural certificate about H_{ont} . The underdetermination itself has measurable structure.

2 Review of the Projection Fiber Framework

We briefly recapitulate the essential definitions from Paper I [1]. The reader is referred to that paper for full proofs and Z3 verification details.

2.1 Projection and Fibers

Let T be a formal theory with model class $\mathcal{M}(T)$. A projection map $\Pi : \mathcal{M}(T) \rightarrow \mathcal{M}(T_\sigma)$ extracts from each model its constraint shadow — the interpretation of a reduced signature Σ . The fiber over a constraint model m is the preimage:

$$\text{Fib}(m) = \Pi^{-1}(m) = \{M \in \mathcal{M}(T) : \Pi(M) = m\}$$

If Π is non-injective ($|\text{Fib}(m)| > 1$ for some m), then the fiber contains ontologically distinct states that the projection conflates.

2.2 Invariance and Independence

A predicate P on $\mathcal{M}(T)$ is projection-invariant if all models within the same fiber agree on P :

$$\forall M_1, M_2 : \Pi(M_1) = \Pi(M_2) \Rightarrow (P(M_1) \leftrightarrow P(M_2))$$

The central result of Paper I: a predicate P is independent of the projected theory T_σ if and only if P is non-invariant on the fibers of Π . This was machine-verified as a Z3 theorem (the ‘independence iff non-invariance’ biconditional).

2.3 The Epistemic Boundary

Paper I further established:

(1) Fiber structure: fibers carry metrics (positive separation between distinct fiber-mates), dynamics (fiber-preserving evolution), and admissible selection principles (action functionals satisfying structural constraints).

(2) Epistemic Boundary Theorem: multiple admissible variational principles can produce the same observable outcome. The specific action is underdetermined from the projection side.

(3) Arrow Underdetermination Theorem: the thermodynamic arrow and metric character of hidden dynamics are not projection-determined.

These results characterize the geometry of epistemic limitation. The present paper complements them with a constructive bound on ontological complexity.

3 Independence Counting and Fiber Cardinality

The central observation is elementary but powerful: each logically independent predicate detected from the observation side forces a structural split in the fiber.

3.1 The Binary Split Principle

Let P be a predicate that is non-invariant on the fibers of Π . By definition, there exist fiber-mates M_1, M_2 with $\Pi(M_1) = \Pi(M_2)$ and $P(M_1) \neq P(M_2)$. The fiber therefore contains at least two distinct points: one satisfying P and one satisfying $\neg P$.

Now let Q be a second predicate, also non-invariant, and orthogonal to P — meaning the truth value of Q is not determined by the truth value of P . Then the fiber must contain models realizing all four truth-value combinations:

$$(P, Q) \in \{(\top, \top), (\top, \perp), (\perp, \top), (\perp, \perp)\}$$

If any combination were absent, it would imply a logical dependence between P and Q within the fiber, contradicting their orthogonality.

In general, n pairwise orthogonal non-invariant predicates force:

$$|\text{Fib}(o)| \geq 2^n$$

This is an exponential lower bound on fiber cardinality, computable from the number of detected independent predicates.

3.2 Orthogonality and Pairwise Independence

Two non-invariant predicates P and Q are orthogonal if knowing the truth value of P does not constrain the truth value of Q within the fiber. Formally: for each truth value v of P , both $Q = \top$ and $Q = \perp$ are realized among fiber-mates satisfying $P = v$.

In the Cantor model, the Continuum Hypothesis (CH: $\mathfrak{c} = \aleph_1$) and the Generalized Continuum Hypothesis at $\aleph_1$ (GCH: $2^{\aleph_1} = \aleph_2$) are orthogonal. This is because:

- CH is about the relationship between $\aleph_0$ and $\mathfrak{c}$.
- GCH at $\aleph_1$ is about the relationship between $\aleph_1$ and $2^{\aleph_1}$.

These concern different levels of the cardinal hierarchy. Consistency results (from forcing) show that all four combinations are realizable within ZFC. Our Z3 verification confirms this: four models with truth-value combinations $(\top, \top)$, $(\top, \perp)$, $(\perp, \top)$, $(\perp, \perp)$ all project to the same constraint shadow.

4 The Fiber Dimension Theorem

We now state and verify the main result.

4.1 Statement

Let $\Pi : H_{\text{ont}} \rightarrow H_{\text{obs}}$ be a non-injective projection with fibers $\text{Fib}(o) = \Pi^{-1}(o)$. Let $P_1, \dots, P_n$ be predicates on H_{ont} such that:

- (1) Each P_i is non-invariant on the fibers of Π (i.e., independent of the projected theory).
- (2) The P_i are pairwise orthogonal (truth values vary independently within fibers).

Then:

$$|\text{Fib}(o)| \geq 2^n$$

The proof is by construction: the 2^n truth-value combinations of n binary predicates must each be realized by at least one distinct fiber-mate, or orthogonality would be violated.

4.2 Z3 Verification

We verify the theorem for $n = 2$ in the Cantor shadow theory using the Kleis verification language and Z3. The `pot_bridge.kleis` source file encodes a structure `FiberDimensionBound` with:

Four model elements: `quad_TT`, `quad_TF`, `quad_FT`, `quad_FF`, each tagged with an `OntTag` value.

Fiber membership: all four project to the same observable (`obs_eq(project(q), project(model_A))`) for each q .

Predicate assignments: the first predicate (CH, encoded as `distinguisher`) is true on `quad_TT` and `quad_TF`, false on `quad_FT` and `quad_FF`. The second predicate (GCH at $\aleph_1$, encoded as `distinguisher2`) is true on `quad_TT` and `quad_FT`, false on `quad_TF` and `quad_FF`.

Pairwise distinctness: all six pairs are asserted distinct via `¬ ont_eq`.

Example	Result	Z3 Time
Two independent predicates require four fiber points	SAT	< 1 ms
Second predicate is also non-invariant on fibers	SAT	< 1 ms
Predicates are orthogonal: independent of each other	SAT	< 1 ms
Orbit bound: consecutive distinct states from evolution	SAT	< 1 ms
FIBER DIMENSION THEOREM: observable lower bound	SAT	< 1 ms
OBSERVABLE LOWER BOUND: detection computable from H_{obs}	SAT	< 1 ms

Table 1: Z3 verification results for the Fiber Dimension Theorem (6 examples, all SAT).

4.3 Observability of the Bound

The critical feature of this lower bound is that it is computable from H_{obs} alone. The detection procedure is:

- (1) Choose a candidate predicate P on H_{ont} .
- (2) Ask Z3: is $P(m_1) \wedge \neg P(m_2) \wedge \text{obs_eq}(\Pi(m_1), \Pi(m_2))$ satisfiable?
- (3) If SAT: P is non-invariant. Increment the independence count. Double the lower bound.
- (4) Repeat for additional predicates.

No access to H_{ont} is required — only the projection map and the axiom system. The solver does the structural certification. Each SAT result is a proof certificate that the fiber is at least twice as large as previously known.

5 Orbit Bounds from Fiber Dynamics

A second, independent source of lower bounds comes from fiber dynamics.

5.1 The Orbit Bound

Let $\text{fiber_evolve} : \text{OntTag} \rightarrow \text{OntTag}$ be a fiber-preserving evolution operator (i.e., $\text{obs_eq}(\Pi(s), \Pi(\text{fiber_evolve}(s)))$ for all s). If fiber_evolve produces k consecutive distinct states starting from a seed s :

$$s, \text{fiber_evolve}(s), \text{fiber_evolve}^2(s), \dots, \text{fiber_evolve}^{k-1}(s)$$

with $\text{fiber_evolve}^i(s) \neq \text{fiber_evolve}^j(s)$ for consecutive pairs, then the fiber contains at least k distinct points: $|\text{Fib}(o)| \geq k$.

In the Cantor model, the FiberDynamics structure from Paper I establishes that fiber_evolve produces two consecutive distinct states (the `non_trivial` and `iterable` axioms), giving $k \geq 2$. This orbit bound is weaker than the predicate bound ($2 < 4$) but is derived from a completely different mechanism — dynamics rather than logic — providing independent confirmation of non-trivial fiber structure.

5.2 Combined Estimates

The predicate bound and orbit bound are independent: one comes from the logic of independent statements, the other from the dynamics of fiber-preserving evolution. The combined lower bound is:

$$|\text{Fib}(o)| \geq \max(2^n, k)$$

where n is the number of detected independent predicates and k is the orbit length. For the Cantor model: $\max(2^2, 2) = \max(4, 2) = 4$.

The combined bound can only grow as more independent predicates are detected or longer orbits are found. It is a monotonically non-decreasing certificate of ontological complexity.

6 The Cantor Instantiation

We now ground the abstract theorem in the concrete case of Cantor’s cardinal arithmetic.

6.1 CH and GCH as Orthogonal Predicates

The Continuum Hypothesis (CH: $\mathfrak{c} = \aleph_1$) and the Generalized Continuum Hypothesis at $\aleph_1$ (GCH: $2^{\aleph_1} = \aleph_2$) are our two independent predicates.

CH concerns the first level of the cardinal hierarchy: is the continuum the first uncountable cardinal? GCH at $\aleph_1$ concerns the second level: does the power set of $\aleph_1$ equal the second uncountable cardinal? These are logically independent questions about different levels of the transfinite.

In the Kleis formalization, CH is encoded as the existing distinguisher predicate from Paper I, and GCH at $\aleph_1$ as the new distinguisher2 predicate. The structure `FiberDimensionBound` introduces four model elements — `quad_TT` (both true), `quad_TF` (CH true, GCH false), `quad_FT` (CH false, GCH true), `quad_FF` (both false) — all in the same projection fiber, all pairwise distinct.

Z3 confirms:

- (1) All four models project identically (same observable fiber).
- (2) All six pairwise distinctness assertions hold.
- (3) The two predicates vary independently: knowing CH’s truth value does not constrain GCH’s truth value within the fiber, and vice versa.
- (4) The combined system with the fiber dynamics orbit, the admissible action, and the epistemic boundary remains consistent.

This gives a concrete, machine-verified instance: the Cantor shadow theory certifies $|\text{Fib}| \geq 4$ from two independent predicates, each detectable from the constraint (projection) side alone.

6.2 Scaling: More Predicates, Larger Bounds

The Cantor shadow theory admits further independent predicates beyond CH and GCH at $\aleph_1$:

- GCH at $\aleph_2$: whether $2^{\aleph_2} = \aleph_3$
- GCH at $\aleph_n$ for each n : an infinite family of independent predicates
- Suslin’s Hypothesis, Martin’s Axiom, and other set-theoretic principles known to be independent of ZFC

Each additional independent predicate that can be detected from the shadow doubles the lower bound. In principle, n independent GCH-type predicates give $|\text{Fib}| \geq 2^n$, growing exponentially. The cardinal hierarchy itself provides an inexhaustible supply of independent predicates, suggesting that the fiber dimension of the Cantor model is not merely finite but grows without bound.

This is consistent with the set-theoretic fact that the class of ZFC models is a proper class — uncountably many models exist, each potentially differing on uncountably many independent statements. The Fiber Dimension Theorem captures a shadow of this richness as a computable lower bound.

7 Discussion

We address the significance, limitations, and future directions of the Fiber Dimension Theorem.

7.1 The Epistemic Inversion

The companion paper established a series of negative results: the specific action is underdetermined, the thermodynamic arrow is hidden, the flow character is inaccessible. These are limits on what observers in H_{obs} can know.

The Fiber Dimension Theorem inverts this direction. It says: from exactly the same observational data that reveals epistemic limits, one can extract positive structural information about H_{ont} . Each independence result detected from the projection side is simultaneously:

- (a) A negative certificate: the predicate cannot be determined from observations.
- (b) A positive certificate: the fiber must be at least twice as large.

The underdetermination itself has measurable structure. The more you discover you cannot know, the more you can certify must exist. This is not a paradox but a structural feature of non-injective projections: information loss (negative) implies redundancy (positive), and redundancy means ontological multiplicity.

7.2 Implications for Projected Ontology Theory

For POT, the Fiber Dimension Theorem provides the first quantitative tool for bounding the complexity of H_{ont} from H_{obs} .

A key distinction must be made. The abstract ontological space H_{ont} may be infinite-dimensional: the class of all ZFC models is a proper class, the space of all possible physical flows is a function space. No finite bound constrains the abstract space. However, any concrete realization of H_{ont} —

any proposal of the form ‘ $H_{\text{ont}} = \mathbb{C}^k$ ’ or ‘ $H_{\text{ont}} = \mathbb{R}^n$ ’ — commits to a finite-dimensional codomain. It is this codomain that the Fiber Dimension Theorem constrains from below.

The fundamental decomposition is rank-nullity applied to the projection:

$$\dim(H_{\text{ont}}) = \dim(\ker(\Pi)) + \dim(H_{\text{obs}})$$

The observable universe is the image of Π ; the fiber is the kernel; the ontological space is both. The Fiber Dimension Theorem gives a lower bound on the kernel: $\dim(\ker(\Pi)) \geq 2^n$. Therefore, any finite-dimensional codomain proposed for H_{ont} must satisfy:

$$\dim(\text{codomain}) \geq \dim(H_{\text{obs}}) + 2^n$$

This is a falsifiability criterion. If H_{obs} has effective dimension d (for spacetime observations, $d = 4$), then the smallest concrete ontological space must have dimension at least $d + 2^n$. With $n = 1$ independent predicate: $\dim \geq 6$. With $n = 2$: $\dim \geq 8$.

This constraint applies directly to existing POT formalizations. In the Kleis source files for Projected Ontology Theory [6], the ontological Hilbert space is concretely modeled as $H_{\text{ont}} = \mathbb{C}^3$ (the type `Vec3C`, representing three complex modal amplitudes (ψ_1, ψ_2, ψ_3)), while the observable space is $H_{\text{obs}} = \mathbb{R}^4$ (the type `Vec4R`, representing spacetime fields). The Green kernel G projects from $\mathbb{C}^3$ (real dimension 6) to $\mathbb{R}^4$ (dimension 4), leaving a kernel of real dimension $6 - 4 = 2$. This kernel can support at most $n = 1$ independent fiber direction, since $2^1 = 2$.

The Fiber Dimension Theorem therefore makes a concrete prediction about this formalization: if a second independent predicate is detected from the observation side — a phenomenon not determined by the spacetime projection that is orthogonal to the first — then $\mathbb{C}^3$ is too small. The ontological codomain must be enlarged. Conversely, $\mathbb{C}^4$ (real dimension 8) would support $n = 2$, and $\mathbb{C}^6$ (real dimension 12) would support $n = 3$. Each additional independent predicate detected from H_{obs} raises the dimensional floor.

The abstract H_{ont} is infinite-dimensional and unconstrained. But the question ‘what is the smallest concrete space that can host the ontological dynamics while respecting what observations force?’ has a finite, computable, and testable answer. That answer grows every time a new independent predicate is detected.

In set theory: the number of known independent statements of ZFC provides a logarithmic estimate of the model-theoretic complexity of the fiber. The supply of independent statements in set theory is inexhaustible, suggesting that no finite-dimensional codomain suffices.

In control theory: the number of unobservable modes gives the fiber dimension directly (via the Kalman decomposition). The Fiber Dimension Theorem generalizes this to non-linear, non-smooth projections.

The practical implication for POT: rather than speculating about the richness of H_{ont} , one can compute a lower bound on its codomain dimension from the data. The ontology is at least as rich as the observations force it to be.

7.3 Limitations

The Fiber Dimension Theorem provides a lower bound, not an exact count. The actual fiber cardinality may be much larger (and in the set-theoretic case, is a proper class). The bound is tight only when all independent predicates have been enumerated, which in general is undecidable.

The orthogonality condition (pairwise independence of predicates) must be verified for each pair, which requires $O(n^2)$ Z3 queries for n predicates. In practice, domain knowledge often provides orthogonality for free (e.g., GCH at different levels concerns different cardinals).

The orbit bound from dynamics is limited by the axioms: the current formalization proves only consecutive distinctness, not full cycle detection. Stronger axioms (e.g., no-return conditions) would tighten this bound.

Finally, the theorem counts distinct fiber points but does not characterize their topological or geometric relationship. The fiber metric from Paper I provides some geometric information, but a full characterization of fiber topology remains open.

8 Conclusion

We have derived the Fiber Dimension Theorem: a computable lower bound on the cardinality of projection fibers — and hence on the ontological degrees of freedom — from observable independence results. The theorem is:

$$|\text{Fib}(o)| \geq 2^n$$

where n is the number of pairwise orthogonal, independently non-invariant predicates detected from the projection side.

This result completes the epistemic picture begun in Paper I. Together, the two papers establish:

- (1) What cannot be known: the specific action, the thermodynamic arrow, the flow character of ontological dynamics (Paper I).
- (2) What must exist: a minimum number of ontologically distinct states, growing exponentially with the number of detected independent predicates (this paper).

The key insight is that the same observational data that reveals epistemic limits simultaneously certifies ontological complexity. Independence results are dual-use: they prove underdetermination and they prove existence.

For the Cantor case study, two independent predicates (CH and GCH at $\aleph_1$) force $|\text{Fib}| \geq 4$, verified by Z3 with all 33 `pot_bridge.kleis` examples passing. The framework extends naturally: each new independent predicate doubles the bound, and the cardinal hierarchy provides an inexhaustible supply.

The projection determines not the content of the hidden layer, but a lower bound on its complexity. That lower bound is the constructive content of underdetermination.

8 References

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Appendix

A Kleis Source Files

The complete verified source files are:

pot_bridge.kleis — The projection-fiber bridge. 10 structures (including the new FiberDimensionBound), 33 verified examples covering: non-injective projection, kernel distinguishers, multi-model fibers, the invariance biconditional, fiber dynamics and metrics, admissible action, epistemic boundary, flow predictions, and the Fiber Dimension Theorem. Total verification time: approximately 7 seconds.

cantor_set_theory.kleis — The Cantor shadow theory. 12 structures, 35 axioms, 19 verified examples. Total verification time: approximately 25 seconds.

To reproduce all results:

```
kleis test examples/cantor/pot_bridge.kleis
```

```
kleis test examples/cantor/cantor_set_theory.kleis
```

All 52 examples (33 + 19) should pass with exit code 0.